

## 第二章 函数的导数(Derivative)

## 2.1 导数的定义(The Definition of Derivative)



## Multiple Choice Questions

1. Which of the following values is the average rate of change of  $f(x) = \sqrt{x+2}$  over the interval  $[-1, 2]$ ?

(A) -4

(B)  $-\frac{1}{3}$ (C)  $\frac{1}{3}$ 

(D) 3

2. Let  $f(x) = 4 - 3x$ . Which of the following is equal to  $f'(0)$ ?

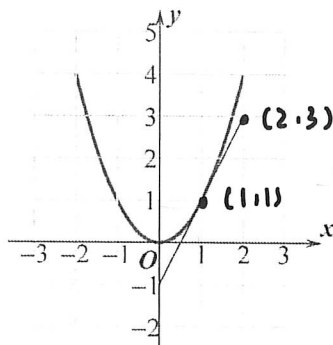
(A) 4

(B) -4

(C) -3

(D) 4

3. The graph of  $f$  and its tangent line at  $x=1$  is shown below, find  $f'(1) = (\quad)$ .



(A) 0

(B)  $\frac{1}{2}$ 

(C) 1

(D) 2

4. Let  $f(x) = kx^3 - 2x$ , ( $k \neq 0$ ). The average rate of  $f(x)$  over the interval  $[-2, -1]$  is 3, find  $k$ .

(A)  $\frac{5}{7}$ (B)  $\frac{1}{7}$ (C)  $-\frac{1}{7}$ (D)  $-\frac{5}{7}$ 

5. **Calculator** Let  $y = \tan e^x - x^2 + 1$ . Find  $\frac{dy}{dx} \big|_{x=0.3}$ .

(A) -12.056

(B) 0

(C) 271.066

(D) 273.066

6. **Calculator** The graph of  $y = \sqrt{x^2 + 1} - 3x$  crosses the  $x$ -axis at one point in the interval  $[0, 1]$ . What is the slope of the graph at this point?

(A) 0.354

(B) -0.001

(C) -2.666

(D) 0

7. The population of insects at time  $t$  days is modeled by the function  $P$  with first derivative  $P'(t) = 0.1t^2 + 10t + 120$ . Which of the following statements is true?

- (A) At time  $t = 10$ , the population of insects is 230.  
 (B) At time  $t = 10$ , the population of insects is increasing at a rate of 230 insects per day.  
 (C) At time  $t = 10$ , the population of insects is decreasing at a rate of 230 insects per day.  
 (D) At time  $t = 10$ , the rate of change of the number of insects is increasing at a rate of 230 insects per day.

8. If the line tangent to the graph of a function  $g$  at the point  $(1, 4)$  passes through the point  $(-2, -5)$ , what is the slope of this line?

- (A) 3 (B) -3 (C)  $\frac{1}{3}$  (D)  $-\frac{1}{3}$



### Free Response Questions

9. Let  $f(x) = x^2 - 2x$

(a) Find the average rate of change of  $f$  over  $[1, 3]$ .

(b) Find  $\frac{f(2+h) - f(2)}{h}$ .

(c) Find the instantaneous rate of change of  $f$  at  $x = 2$ .

$$(a) \frac{\Delta y}{\Delta x} = \frac{(9-6)-(1-2)}{3-1} = \frac{4}{2} = 2$$

$$(b) \frac{h^2 + 2h + 0}{h} = h + 2$$

$$(c) \lim_{h \rightarrow 0} (h + 2) = 2$$

10. Use the definition of derivative to find the derivative of  $f(x) = x^3$ , then find  $f'(2)$ .

$$\begin{aligned} f(x) &= x^3 \\ \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h} &= \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h} \\ &= \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3}{h} \\ &= \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2) \\ &= 3x^2 \\ f'(2) &= 12 \end{aligned}$$